Algorithm Sample 3

Dataset-expansion sample

Procedure

The pseudocode below is drawn from a source-backed image-to-LaTeX benchmark and reproduced verbatim.

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1 **Initialize** actor network μ_θ and critic networks Q_{ω_1} and Q_{ω_2}.

2 **Initialize** corresponding target networks: $\theta' \leftarrow \theta$, $\{\omega'\}_1 \leftarrow \omega_1$, $\{\omega'\}_2 \leftarrow \omega_2$ and choose $\rho_{\mathrm{p}} \in (0, 1)$.

3 **Initialize** replay buffer $\mathcal{R}$.

4 **Choose** a batch size K, a gradient based optimization algorithm and a corresponding learning rate $\lambda_{\text{actor}}, \lambda_{\text{critic}} > 0$ for both optimization procedures, a time step size Δt and a stopping criterion.

5 **Choose** standard deviation exploration noise σ_{expl} and lower and upper action bounds $a_{\text{low}}, a_{\text{high}}$.

6 Repeat

7 **Select** clipped action and step the environment dynamics forward.

$$a = \operatorname{clip}\big(\mu_\theta(s) + \varepsilon, a_{\text{low}}, a_{\text{high}}\big), \quad \varepsilon \sim \mathcal{N}\big(0, \sigma_{\text{expl}}\big).$$

8 **Observe** next state s', reward r, and done signal d and store the tuple (s, a, r, s', d) in the replay buffer.

9 If s' **is terminal** then

10 **Reset** trajectory.

11 End If

12 For j **in range**(*update frequency*) do

13 **Sample** batch $\mathcal{B} = \left\{\left(s^{(k)}, a^{(k)}, r^{(k)}, \{s'\}^{(k)}, d^{((k)\}}\right)\right\}_{k=1}^{K}$ from replay buffer.

14 **Compute** targets (Clipped Double Q-learning and policy smoothing).

$$y(r, s', d) = r + (1 - d)\min_{i=1,2}\left\{Q_{\{\omega'\}_i}(s', \tilde{a})\right\}, \quad \tilde{a} = \operatorname{clip}\big(\mu_{\theta'}(s) + \varepsilon, a_{\text{low}}, a_{\text{high}}\big), \quad \varepsilon \sim \mathcal{N}\big(0, \sigma_{\text{target}}\big).$$

15 **Estimate** critic gradient $\nabla_\omega L(Q_\omega^{\mu_\theta})$ by

$$\nabla_{\omega_i}\left(\tfrac{1}{K}\textstyle\sum_{k=1}^{K}\left(Q_{\omega_i}\big(s^{(k)}, a^{(k)}\big) - y\Big(r^{(k)}, \{s'\}^{(k)}, d^{(k)}\Big)\right)^2\right), \quad \text{for } \; i = 1, 2.$$

16 **Update** the critic parameters ω_i based on the optimization algorithm.

17 If $j \bmod$ *policy delay frequency* $= 0$ then

18 **Estimate** actor gradient $\nabla_\theta J(\mu_\theta)$ by

$$\nabla_\theta\Big(\tfrac{1}{K}\sum_{k=1}^{K} Q_{\omega_1}\big(s^{(k)}, \mu_{\theta(s^{(k)})}\big)\Big).$$

Update the actor parameters θ based on the optimization algorithm.

Update target networks softly:

$$\theta' \leftarrow \rho_{\mathrm{p}}\theta' + \big(1-\rho_{\mathrm{p}}\big)\theta, \quad \{\omega'\}_i \leftarrow \rho_{\mathrm{p}}\{\omega'\}_i + \big(1-\rho_{\mathrm{p}}\big)\omega_i, \quad \text{for} \;\; i = 1, 2.$$

End If

End For

Until stopping criterion is fulfilled.